Mouhamed Nourou Dine Cisse

AI Engineer | Machine Learning Engineer | Data Scientist

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PROFESSIONAL SUMMARY

Applied Mathematics and Computer Science engineering student at ENSIIE, pursuing a double degree in Artificial Intelligence (M2IA) at Universite Paris-Saclay. Experience building and evaluating machine learning models, multimodal and NLP systems, data pipelines, and LLM-powered applications. Seeking a six-month end-of-studies internship starting in March 2027 in AI engineering, machine learning, or data science.

TECHNICAL SKILLS

Programming and data: Python, SQL, R, C++, Java, JavaScript, Bash, Pandas, NumPy, data analysis, ETL/ELT.

Machine learning: PyTorch, scikit-learn, PyTorch Geometric, Statsmodels, supervised and unsupervised learning, feature engineering, model validation and evaluation, time-series forecasting.

AI: Deep learning, Transformers, NLP, computer vision, multimodal AI, large language models, retrieval-augmented generation (RAG), AI agents, reinforcement learning, graph neural networks.

Engineering and tools: FastAPI, LangGraph, Hugging Face, Streamlit, PostgreSQL, Oracle, Docker, Git, Linux, Azure.

PROFESSIONAL EXPERIENCE

Applied AI and Data Engineering Intern | Prodigious Groupe, Publicis, Paris | May-July 2026

- Built data pipelines and a modular Python/FastAPI platform for operational-data workflows, using typed schemas, services, repositories, and database integrations.
- Developed a LangGraph supervisor with specialized workers for structured-data queries, business workflows, and retrieval-augmented generation over internal documentation.
- Evaluated 27 scenarios twice: all 54 technical runs succeeded, and 48 of 54 outputs met the functional rubric (88.9%).

Machine Learning Intern | Bangath Systems and Gainde 2000, Paris and Dakar | May-August 2025

- Developed predictive models on 110,260 Port of Dakar records: Random Forest achieved 0.96 F1-score for irregularity detection; SARIMA achieved 0.67 MAE for traffic forecasting.
- Prepared 136 variables using SQL/PLSQL and deployed FastAPI inference services integrated with Oracle APEX.

SELECTED PROJECTS

Autoregressive Decoder-Only Transformer from Scratch | PyTorch | 2026

Implemented a compact language-model decoder with causal multi-head attention, Rotary Position Embeddings (RoPE), RMSNorm, SwiGLU feed-forward layers, and key-value caching. Added autoregressive decoding, training and perplexity evaluation on TinyStories.

Auditing Vision-Language Model Robustness to Face Accessories | Universite Paris-Saclay | 2026

Conducted a paired evaluation of MiniCPM-V 4.6 and SigLIP 2 on 720 synthetic identities, comparing portraits with and without glasses, masks, and caps. Measured age, gender, and ethnicity accuracy, job-suitability rating shifts, and demographic subgroup effects; all 18 objective accuracy comparisons declined with accessories, 13 significantly.

Nash-DQN for Multi-Agent Trading | McMaster University | 2026

Implemented a PyTorch multi-agent trading environment and Deep Q-Network components to study Nash-equilibrium learning, using limit-order-book dynamics and a TWAP baseline.

Neural NLP and Named Entity Recognition | ENSIIE | 2025-2026

Explored Word2Vec and FastText embeddings; implemented BiLSTM-CNN for named entity recognition and CNN text classification; fine-tuned Transformer models for supervised NLP tasks.

Graph Neural Networks for Malware Detection | Sorbonne Data Challenge | 2025

Built a PyTorch Geometric graph-classification pipeline using control-flow graphs for malware detection.

ETLia | AI-Assisted Data Integration | 2025

Co-founded an AI platform concept for automating ETL workflows through natural-language interaction; awarded First Prize at the ENSIIE Entrepreneurship Challenge.

EDUCATION

ENSIIE and Universite Paris-Saclay | Engineering Degree in Applied Mathematics and Computer Science | 2024-2027
Double degree: Master 2 in Artificial Intelligence (M2IA), Universite Paris-Saclay, in progress (2026-2027).

M2IA coursework: Data Readiness and Evaluation in AI Systems; NLP Today; Deep Learning for NLP; Information Retrieval 2: Recent Advances; Reinforcement Learning; Probabilistic Generative Models; Advanced Optimization and Automated ML; Scientific Machine Learning; Signal Processing M2 (chosen option).

McMaster University | Academic Exchange, Hamilton, Canada | Winter 2026
Completed coursework: Neural Networks and Machine Learning; Applied Artificial Intelligence; Generalized Linear Models; Time Series; Financial Markets and Derivatives.

CPGE MP* | Lycee Albert Schweitzer, Mulhouse | 2022-2024

Preparatory program in advanced mathematics, physics, and computer science.

CERTIFICATION AND AWARDS

NVIDIA Certificate of Competency: Enhancing Data Science Outcomes With Efficient Workflow | April 2026.

First Prize, ENSIIE Entrepreneurship Challenge 2025: Co-founded ETLia. Ranked first in Data Analysis at ENSIIE.

LANGUAGES

French: Native | English: C2 | Arabic: B2 | German: B1